

A proof of the conjectured linear recurrence and generating function for OEIS A252689

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1 The conjecture

The Online Encyclopedia of Integer Sequences records [A252689](#) as

Number of $(n+2) \times (2+2)$ 0..3 arrays with every consecutive three elements in every row and column having exactly two distinct values, and in every diagonal and antidiagonal not having exactly two distinct values, and new values 0 upwards introduced in row major order

and carries in its Formula section the lines:

Empirical: $a(n) = 5a(n-1) - 4a(n-2) - 11a(n-3) + 9a(n-4) + 12a(n-5) + 12a(n-6) - 34a(n-7) - 8a(n-8) + 28a(n-9) - 8a(n-10)$ for $n > 11$.

Empirical g.f.: $x(96 - 338x - 122x^2 + 1123x^3 - 34x^4 - 626x^5 - 1907x^6 + 1306x^7 + 1728x^8 - 1488x^9 + 288x^{10}) / ((1-x)(1+x)(1-2x)(1-3x+x^2)(1-2x^2)(1-2x^3))$. - Colin Barker, Dec 05 2018

That line carries no separate signature; the entry itself is by R. H. Hardin (Dec 20 2014).

The line quoted ending in that signature is contributed by Colin Barker on Dec 05 2018.

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 8, last modified Dec 05 2018, the entry records no proof and links to none, so the statement is open. This note settles it.

Written out, the claim is

$$a(n) = 5a(n-1) - 4a(n-2) - 11a(n-3) + 9a(n-4) + 12a(n-5) + 12a(n-6) - 34a(n-7) - 8a(n-8) + 28a(n-9) - 8a(n-10).$$

stated for $n > 11$.

Written out, the claim is

$$\sum_{n \geq 1} a(n) x^n = \frac{96x - 338x^2 - 122x^3 + 1123x^4 - 34x^5 - 626x^6 - 1907x^7 + 1306x^8 + 1728x^9 - 1488x^{10} + 288x^{11}}{1 - 5x + 4x^2 + 11x^3 - 9x^4 - 12x^5 - 12x^6 + 34x^7 + 8x^8 - 28x^9 + 8x^{10}}.$$

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let A be an $n' \times 4$ array over $\{0, \dots, 3\}$ with $n' = n + 2$ rows.

Every three such cells, with $(\delta, \varepsilon) \in \{(0, 1), (1, 0)\}$, must carry exactly 2 distinct values.

Every three such cells, with $(\delta, \varepsilon) \in \{(1, -1), (1, 1)\}$, must not carry exactly 2 distinct values.

The entry's labelling clause says that reading the array row by row the values $0, \dots, 3$ first occur in increasing order.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Every condition looks at three cells in a straight line, so it spans at most three consecutive rows. Taking as state the last two rows together with a counter for how many values have been introduced makes every condition decidable at the moment its lowest row is read.

An array of n' rows is then exactly a walk of length n' from the empty state, so $a(n) = w^\top M^{n'} f$ and the count is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 672 states, 672 after trimming; the forward identification of states with equal numbers of completions of every length, then the mirror identification on the edges coming in, leaves $S = 20$ blocks, and the count satisfies the linear recurrence given by the characteristic polynomial of that 20×20 matrix. The bound is computed, not assumed.

5 The decision procedure

Let $c_1, \dots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^D c_i a(n-i)$, let $q(t) = t^D - \sum_{i=1}^D c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m = \tilde{w}^\top L^{m-1} q(L) \tilde{f} \quad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a , one gets $u_m = a(n) - \sum_{i=1}^D c_i a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^\top L^{m-1} y$ with $y = q(L) \tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L , of degree 20: writing that polynomial as $t^{20} - \sum_{i=1}^{20} \lambda_i t^{20-i}$ gives $u_{m+20} = \sum_{i=1}^{20} \lambda_i u_{m+20-i}$. Hence 20 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.
4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.